

The Pacific growth spurt, and a new approach to measuring the economic performance of the Pacific islands*

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Abstract

Gross National Disposable Income (GNDI) is a better measure of average living standards than either Gross National Income (GNI) or Gross Domestic Product (GDP), but it is rarely used. The Pacific region is where measuring economic performance using GNDI matters most, and there it matters a lot. Using GNDI changes our perspective on the economic performance of several Pacific Island Countries (PICs) and the region as a whole. It establishes the uniqueness of the PICs from an economic perspective and reveals a growth spurt in the 2010s that is otherwise hidden. Resort to GNDI shows that per capita income growth of the PICs exceeded that of any other region between the global financial crisis and the pandemic. Using GNDI to measure economic performance is becoming more important over time, as the divergence between GNDI and GDP in the Pacific has grown. The paper explains both this increasing divergence and the growth spurt. It also establishes that reliance on GNDI provides a better framework for understanding the economics of the Pacific than either of the two dominant frameworks, namely the “GDP” and the “MIRAB” approaches. It also does not suffer from the ambiguity in its treatment of remittances to which GNI is subject. GNDI should be used routinely in any analysis of the Pacific. However, the treatment of foreign aid in the balance of payments needs to be improved in some countries, while some others need to report balance of payments data more promptly.

Keywords: Pacific, small states, economic growth, national accounts, aid, remittances.

JEL Classifications: F2; F21; F35; F43; O11.

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Acronyms

BOP	Balance of Payments
COO	Cook Islands
CPI	Consumer Price Index
FDI	Foreign Direct Investment
FJI	Fiji
FSM	Federated States of Micronesia
GDP	Gross Domestic Product
GFC	Global Financial Crisis
GNDI	Gross National Disposable Income
GNI	Gross National Income

IMF	International Monetary Fund
KIR	Kiribati
MHL	Marshall Islands
NFI	Net Foreign Income
NPI	Net Primary Income
NRU	Nauru
NSI	Net Secondary Income
ODA	Official Development Assistance
OECD	Organisation for Economic Co-operation and Development
PICs	Pacific Island Countries
PLW	Palau
SLB	Solomon Islands
SNA	System of National Accounts
SWF	Sovereign Wealth Fund
TON	Tonga
TUV	Tuvalu
VDS	Vessel Day Scheme
VUT	Vanuatu
WDI	World Development Indicators
WSM	Samoa

1 Introduction

Gross Domestic Product ([GDP](#)) and Gross National Income ([GNI](#)) are commonly used indicators of economic performance. Gross National Disposable Income ([GNDI](#)) is a superior measure of average living standards. Yet, it is very rarely utilised.

This paper argues that GNDI should be used to assess economic performance in the Pacific. In fact, it shows that using GNDI has become more important over time as the divergence between GNDI and GDP has grown. The paper argues that using GNDI rather than GDP changes our perspective on the economic performance of several Pacific Island Countries ([PICs](#)) as well as the region as a whole. It provides the basis for a better and, at least in the recent past, more optimistic understanding of the economies of the region.

Resort to GNDI shows that, in fact, per capita income growth of the PICs exceeded that of any other region between the global financial crisis and the COVID-19 pandemic. That this incredible growth spurt (average annual per person growth of 4.6 per cent) is missed unless GNDI is used - and can only be explained if GNDI is used - itself makes a compelling case for using GNDI in Pacific economic analysis.

GDP measures the value of all final goods and services produced in an economy. GDP ignores cross-border non-trade income flows, except to the extent that they change domestic production. [GNI](#) partially addresses this shortcoming by augmenting GDP by Net Primary Income ([NPI](#)), which is the net foreign income of (return to) labour, capital, and natural resources. The lesser-known GNDI improves on GNI by including Net Secondary Income ([NSI](#)), which is made up of net unrequited transfers to households and governments. GNDI thus measures the total income available to an economy for final consumption and savings and, hence, is a better measure of

average living standards than either GDP or GNI ([SNA, 2008](#)).

As NPI and NSI are routinely measured and published as part of the Balance of Payments ([BOP](#)), calculating GNDI is simple. Yet, while GNDI is reported in the statistical publications of various PICs, it is completely overlooked in the analysis of Pacific economic performance (as it is almost entirely elsewhere).

The theoretical case for focusing on GNDI is strong. While GDP measures production, GNDI measures national income. As [Stiglitz et al. \(2009\)](#) argue in their landmark report, if a country “wants to know whether its citizens are better-off, ... national income measures are more relevant to this question than GDP”. The assumption is often made that if an income approach is wanted, then GNI should be analysed (see, for example, [Piketty, 2014](#)). However, as [Stiglitz et al. \(2009\)](#) clarify, GNDI “is a more comprehensive aggregate” than GNI and that it “better measures how well off citizens are” (Box 1, p.95).

Despite this strong endorsement, [Stiglitz et al. \(2009\)](#) say nothing more about GNDI in their 292-page report beyond these brief remarks. [Capelli and Vaggi \(2016\)](#) elaborate on the differences between GNDI and GNI to make the case for “Why Gross National Disposable Income Should Replace Gross National Income”, the title of their paper. We build on and improve the [Capelli and Vaggi \(2016\)](#) arguments in several important ways. First, the authors ignore the Pacific, the only region where the use of GNDI really matters. Second, [Capelli and Vaggi \(2016\)](#) focus solely on remittances, which is only one of the sources of divergence between GDP, GNI and GNDI. Third, we make a case against using GNI based on how remittances are actually measured. Fourth, the authors do not calculate GNDI growth rates, which is fundamental to any assessment of economic performance.

Given our measurement concerns about the unreliability of GNI calculations (the third in the previous paragraph), we focus our comparisons exclusively on GDP and GNDI. Section 5 substantiates those concerns and justifies our neglect of GNI.

For most countries, it does not matter whether GDP or GNDI is used for analysis. For 103 out of 171 countries for which data on both are available in 2019, GNDI exceeds GDP by less than 5 per cent. However, it matters for the Pacific. For the 12 PICs this study focuses on, the average difference in 2019 is 40 per cent. When we rank countries by their 2019 ratio of GNDI to GDP, the top four (Kiribati, Tuvalu, the Federated States of Micronesia and the Marshall Islands) are all from the Pacific and are the only ones in the world to have a GNDI/GDP ratio greater than 1.5. In fact, seven of the top 20 countries ranked by GNDI/GDP are from the Pacific region.

One natural explanation is that the PICs are very small countries, and foreign income flows are plausibly more important for small economies. At first glance, Figure 1 lends support to this hypothesis, with a downward-sloping trend line (the red line) indicating a negative correlation between population size and the GNDI/GDP ratio. However, once the PICs are removed from the sample, the GNDI/GDP correlation (the blue line) actually becomes completely insignificant (and positive). The uniqueness of the PIC economies emerges as a clear theme of the analysis of this paper. Some explanations are provided for this, but more research is required.

GNDI is important not only for measuring living standards in the Pacific but also for understanding the economic structure and prospects of the Pacific. Analyses of Pacific economic performance have, over the decades, been divided into two camps. The first “GDP” camp applies to the Pacific, the sort of economic growth analysis used for developing countries in general. It focuses on domestic production and

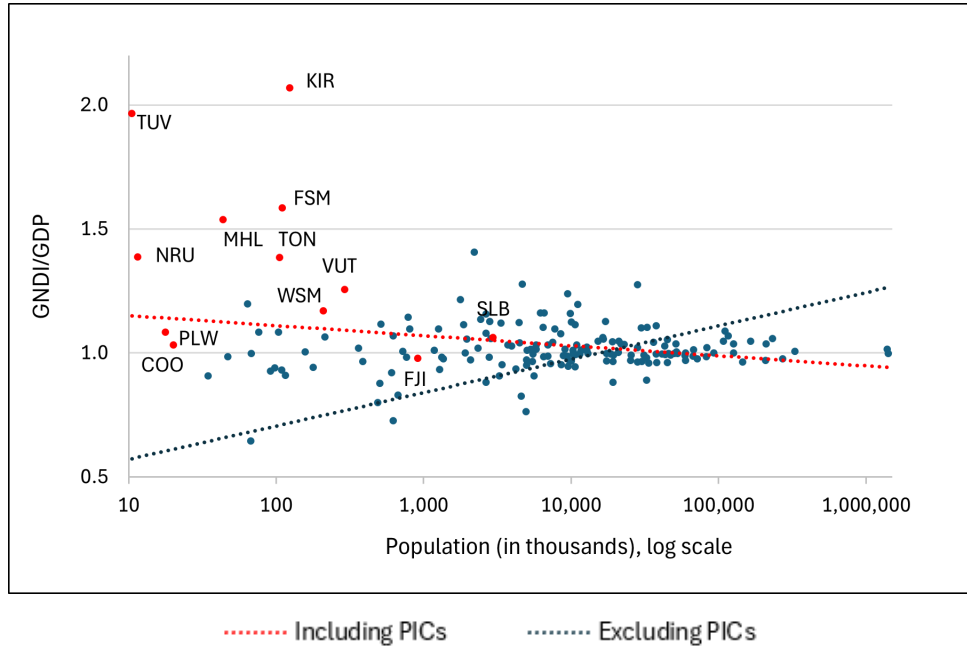


Figure 1: *Correlations of GNDI/GDP ratio and population, 2019.*

Source: [World Bank \(2024\)](#); authors' calculation.

Notes: The sample comprises 171 countries observed for 2019. The PICs are marked red. Data for the Cook Islands is collected from [Cook Islands Statistics Office](#).

Abbreviations: Cook Islands (COO), Fiji (FJI), Federated States of Micronesia (FSM), Kiribati (KIR), Marshall Islands (MHL), Nauru (NRU), Palau (PLW), Samoa (WSM), Solomon Islands (SLB), Tonga (TON), Tuvalu (TUV), and Vanuatu (VUT).

thus on GDP ([Cole and Tambunlertchai, 1993](#)). This camp typically analyses why GDP growth is not faster in the Pacific and puts forward recommendations for its acceleration.

The second “MIRAB” camp emphasises the uniqueness of the Pacific and its reliance on remittances and aid ([Bertram and Watters, 1985](#)).¹ The MIRAB camp is more realistic than the GDP camp in its understanding of Pacific economies, but suffers from two weaknesses. First, MIRAB proponents have never proposed a metric – an

¹‘MIRAB’, a term introduced by [Bertram and Watters \(1985\)](#), stands for ‘migration, remittances, aid and bureaucracy’, the bureaucracy being financed by aid.

alternative to GDP – for measuring Pacific economic performance. Second, MIRAB proponents tend to ignore or under-emphasise financing sources other than aid and remittances, even though these are also important in the Pacific. While other authors have broadened the net to examine other sources of foreign income (see, for example, [Baldacchino, 2006](#)), none of the alternatives to MIRAB provides a systematic way of presenting and comparing different financing sources. GNDI provides a metric for economic performance that can integrate and compare the importance of domestic production, aid, remittances and all other sources of foreign income.

The paper is structured as follows. Section [2](#) sets out data sources and methods. Section [3](#) demonstrates the importance of GNDI in the Pacific. Section [4](#) compares the economic performance of the Pacific using GNDI and GDP. Section [5](#) analyses the different sources of net foreign income that constitute the gap between GNDI and GDP in order to explain both the rapid growth in GNDI and the growing divergence between GNDI and GDP, two phenomena which are, of course, linked. Section [6](#) concludes. Additional data details are in the appendix.

2 Method and data

This section begins with the definitions of GDP, GNI, and GNDI (Section [2.1](#)). It then outlines the components of net primary and secondary income and groups them into five categories that constitute the gap between GDP and GNDI (Section [2.2](#)). It discusses how we measure the growth and volatility of real GDP, GNDI, and its components (Section [2.3](#)), and it presents the sample of countries studied, along with the data sources and gaps (Section [2.4](#)).

2.1 GDP, GNI and GNDI

GDP at purchaser prices represents the total gross value added by all producers operating within a nation's borders, plus taxes on products, minus any subsidies. GDP can be calculated by directly measuring value-added or by using the expenditure approach, which sums the market value of all final goods and services produced in an economy at purchaser prices, subtracting the value of imported goods and services. According to the expenditure method, GDP is defined as:

$$GDP = C + I + G + X - M \quad (1)$$

Where C is household consumption, I is gross capital formation (including fixed capital, inventories, and valuables), G is government consumption, X is the exports of goods and services, and M is the imports of goods and services.

The resident units of an economy not only produce income at home through the production process but also earn foreign primary income. Wages earned by resident workers abroad, dividends and interest earned on foreign assets, and fees paid by foreign companies for access to natural resources, for example, fishing licenses, all contribute to primary income. Conversely, a country must also pay for the services it receives from nonresident units operating within its territory. NPI is the difference between the primary income received by resident units and the primary income paid to nonresident units. By adding NPI to GDP, we arrive at GNI:

$$GNI = GDP + NPI \quad (2)$$

A country's GNI is higher than its GDP, or $GNI/GDP > 1$ if $NPI > 0$.

While the primary income account captures foreign returns, the secondary income account records current *unrequited* transfers between residents and non-residents, that is, those that occur without any corresponding exchange of goods or services. Remittances sent home by migrants, whether to households or, say, churches, as well as aid sent by foreign governments, contribute to secondary income. Conversely, remittances sent abroad by migrant workers must be deducted from the NSI, which is the difference between secondary income received by resident units and secondary income paid to nonresident units.

By combining both NPI and NSI, we arrive at GNDI:

$$GNDI = GDP + NPI + NSI \quad (3)$$

GNDI is larger than GDP ($GNDI/GDP > 1$) when $NPI + NSI > 0$. From Equations (1) and (3), we can derive the relationship between disposable income, savings, and expenditure. Beginning with Equation (3),

$$GNDI = C + I + G + (X - M + NPI + NSI) = C + I + G + CAB \quad (4)$$

where $CAB = (X - M + NPI + NSI)$ is the current account balance. The System of National Accounts (SNA, 2008) defines gross savings (S) as the difference between gross national disposable income and expenditure, $S = GNDI - C - G$. As a result,

$$GNDI = (C + G) + S \quad (5)$$

Thus, GNDI measures the total income available to a country for final consumption and savings. By combining Equations (4) and (5), we can also demonstrate that the

current account balance reflects the savings-investment gap within an economy:

$$CAB = S - I \tag{6}$$

Equations (4) and (6) illustrate the economic significance of GNDI in linking domestic and international transactions. For example, a current account deficit ($CAB < 0$) can be addressed by reducing current domestic expenditures or increasing savings relative to investment.

One way to think of GNDI is that it includes ‘free imports’- those financed by foreigners. Imports are not included in GDP. But if someone else is paying for a country’s imports, those imports should be included in any measure of living standards for that country.

A simple example might help. Consider a country that receives US\$10 million in budget support from foreign governments or remittances from former residents who have migrated abroad. It is all spent on imports. GDP and GNI show no change, but GNDI increases by US\$10 million. Thus, GNDI is the only measure that shows that the country is US\$10 million better off. Now, say the US\$10 million induces an additional US\$1 million of domestic production (since the economy has excess capacity). GDP and GNI increase by US\$1 million, but the economy is better off by US\$11 million, the amount GNDI will increase by. As is clear from these examples, foreign transfers such as aid and remittances are included in GDP only to the extent that they induce a domestic production response.

Ideally, we would measure net domestic product, net national income and net national disposable income by subtracting the consumption of capital (or capital depreciation) from GDP and, by extension, from GNI and GNDI. However, while available,

data on capital depreciation are unreliable, at least for the Pacific (since estimates of capital stock are unavailable) and for many countries as well. We, therefore, conduct the analysis in gross rather than net terms.

2.2 Components of NPI and NSI

The sixth edition of the International Monetary Fund ([IMF](#)) Balance of Payments and International Investment Position Statistics (BPM6) categorises net primary income into three main types: ‘compensation of employees’, ‘investment income’, and ‘other primary income’.

Compensation of employees consists of wages, salaries, and other benefits resident individuals receive from their non-resident employers in exchange for labour. Likewise, it includes wages, salaries, and other benefits resident employers pay to non-resident individuals.

Investment income is the returns from various foreign investments that residents make. This includes foreign direct investment, portfolio investment, and other investments in physical and financial assets.

Other primary income includes revenue from natural and other resources owned by a reporting country but utilised by non-residential institutional units. This category comprises land rented to extract minerals and sub-soil deposits, temporary military bases, and activities related to fishing and forestry. We refer to this category as *returns to non-financial assets*.

Net secondary income is categorised into three main types: current transfers to the ‘general government’, ‘personal transfers’, and ‘other current transfers’.

General government transfers includes all current transfers of goods and services, either in cash or in kind, that the government of the reporting country receives from or pays to the rest of the world without expecting any quantifiable benefit in return. Typically, these transfers include grants from foreign governments and international organisations used to finance current expenditures.

For Pacific economies, foreign aid is the most important type of government transfer. However, there are some differences between government transfers and Official Development Assistance (ODA) in the way it is usually defined. First, ODA is measured by the donors (and reported to and calculated by the Organisation for Economic Co-operation and Development (OECD)), whereas the recipient measures government transfers. Second, ODA includes both current and capital expenditures, whereas government transfers are only current. Third, ODA includes expenditures made outside of the recipient country. These would also be excluded from government transfers. As we show in Section 5, government transfers are about half of the ODA for the Pacific.

Personal transfers comprise all current transfers made between resident and non-resident individuals, whether in cash or kind, regardless of their relationship and the source of the sender's income.

Other current transfers include all types of current transfers that do not take place between individuals and between government entities for the purpose of international cooperation. This category of secondary income includes current international transfers to non-profit institutions serving households, taxes on income and wealth, as well as social contributions made by households to social insurance programs in another country. It also includes gifts, donations, and similar transactions.

As is evident from the above discussion, remittances can take several forms: compensation of employees, part of NPI, personal transfers, part of NSI, and remittances to institutions (e.g., churches) rather than households, which are part of other current transfers, also part of NSI. The World Bank remittances figures equal the sum of the first two. In Section 5, we show that the distinction between employee compensation and personal transfers is not meaningful. This means that GNI cannot be measured accurately (since what is counted as compensation of employees and what is counted as personal transfers is not defined consistently across countries). For this reason, we focus only on GDP and GNDI. Net foreign income (the GNDI-GDP gap) can be decomposed into the following five components:

Aid	=	Current transfers to the general government
Remittances	=	Compensation of employees + Personal transfers
Investment income	=	Income from direct, portfolio, and other investments and central bank reserve assets
Returns on non-financial assets	=	Other primary income
Residual	=	Other secondary income

2.3 Change over time

To examine growth, whether in GDP or GNDI, we need to adjust for inflation. We use the GDP deflator (in local currency) to measure GDP growth. There are two methods to deflate GNDI.

The preferred method proposed by the UN System of National Accounts ([SNA](#)) is to deflate GDP using the GDP deflator, adjust for real terms of trade gains, and then add real NPI and real NSI. Under the SNA, countries can choose the method they want to use to measure real terms of trade gains. They can deflate using either the export price index, import price index, weighted average of the two, or a general price index such as the Consumer Price Index ([CPI](#)) or gross domestic final expenditure. The same general price index is used to deflate NPI and NSI.

The UN also proposes a second approach: deflating nominal GNDI by a single deflator. As the UN says, this approach has the advantage of “a single numeraire, the set of goods and services making up net domestic final expenditures being used throughout”. The UN favours the first method because it argues that real terms of trade gains should be measured using import and/or export price-specific deflators. However, we do not have access to trade-specific deflators (import or export price indices) for all PICs.² Therefore, we utilise the second method. We use the CPI deflator since this is the only general price index available for all PICs.

It turns out that the GDP and CPI deflators often diverge significantly. To show the importance of the deflator, we also show GNDI growth rate using the GDP deflator (denoted by GNDI*) and compare it to GDP growth using the GDP deflator. Since growth is volatile, we calculate the average annual growth rates.

To summarise, we calculate GDP and GNDI per person in constant prices, expressed in local currency, as follows:

$$GDP_{k,t} = \left(\frac{GDP_{c,t}/GDP_{def_t}}{Pop_t} \right) \quad (7)$$

²Seven out of 12 countries studied have data on import and export price indices- FJI, KIR, SLB, TON, TUV, VUT and WSM.

$$GNDI_{k,t} = \left(\frac{GNDI_{c,t}/CPI_t}{Pop_t} \right) \quad (8)$$

$$GNDI_{k,t}^* = \left(\frac{GNDI_{c,t}/GDP_{def_t}}{Pop_t} \right) \quad (9)$$

where subscripts c , k , and t stand for current price, constant price, and year, respectively. $GDP_{def_t} = (GDP_{c,t}/GDP_{k,t})$ is the GDP deflator with $GDP_{def_{2015}} = 1$, CPI is the consumer price index with $CPI_{2015} = 1$, and Pop_t is the current year population.

Volatility is the deviation of a variable around its equilibrium value, which is often referred to as a steady state or trend (Cariolle, 2012). The simplest way to measure volatility is by estimating the standard deviation (or higher order moments) (Ramey and Ramey, 1994). However, the components of net foreign income in Section 2.2 were found to be non-stationary, both in levels and in terms of GDP. Their growth rates are stationary, but some components are close to zero in some years, making some growth rates so large that the calculation of the standard deviation of component-specific growth would be meaningless. An alternative approach is to assess the volatility of the cycle derived from a variable's regression on its temporal trend, as long as the cycle itself is stationary (Pritchett, 2000).

We measure the volatility of national income and its components by estimating the standard deviation of the residuals from the following quadratic³ trend equation:

$$y_t = \alpha + \beta t + \gamma t^2 + \epsilon_t \quad (10)$$

where y_t is national aggregate income (GDP, GNDI and GNDI*) or one of the components of net foreign income (aid, remittances, investment income or return on

³It was found that a quadratic model ensures stationarity of the underlying residuals.

non-financial assets), all expressed in natural logs and in real terms.

The volatility of y_t is measured by estimating the standard deviation of $\tilde{\epsilon}_t$, which represents the cycle of y_t around its trend. The stationarity of $\tilde{\epsilon}_t$ is evaluated by three unit root tests: Augmented Dickey-Fuller or ADF (Dickey and Fuller, 1981), Phillips-Perron or PP (Phillips and Perron, 1988) and Kwiatkowski-Phillips-Schmidt-Shin or KPSS (Kwiatkowski et al., 1992). Where the ADF or PP test fails to reject the null hypothesis of a unit root, we rely on the results of the KPSS test.⁴ The test results are shown in Table A2 of the appendix.

2.4 Countries and data

This study focuses on the 12 independent PICs for which data are available: the Cook Islands (COO), Fiji (FJI), the Federated States of Micronesia (FSM), Kiribati (KIR), the Marshall Islands (MHL), Nauru (NRU), Palau (PLW), Samoa (WSM), the Solomon Islands (SLB), Tonga (TON), Tuvalu (TUV), and Vanuatu (VUT). Papua New Guinea (PNG) is often left out of PIC lists⁵ on account of its size (its population is about twice the global median) and is excluded here for the same reason: our focus is on the small island economies of the Pacific. Australia and New Zealand, which are also island states and members of the Pacific Island Forum, are omitted for the same reasons. Data is not available for Niue, the only other independent Pacific country. Timor-Leste is sometimes included in the analysis of Pacific countries but is excluded here for two reasons: first, it is not a Pacific country;

⁴Empirically, the KPSS test tends to perform better than the ADF and PP tests in detecting mean-reverting behaviour in economic data (see, for example, Hobijn et al., 2004). The KPSS test found 68 out of 70 variables stationary at 5% significance level.

⁵See, for example, the European Commission reports at https://international-partnerships.ec.europa.eu/countries/pacific-islands-countries_en.

second, the unusual treatment of its oil revenue has a large and highly time-varying impact on its GNDI-GDP relationship, making it *sui generis* in nature.⁶

Annual data are collected from two main sources. NPI and NSI data are obtained from the Balance of Payments and International Investment Position Statistics (BOP/IIP) of the IMF. The IMF reports BOP data following the Sixth Edition of its Balance of Payments and International Investment Position Manual (IMF, 2024). All other variables are taken from the World Development Indicators (WDI) of the World Bank (World Bank, 2024). Neither the IMF nor the World Bank reports data on the Cook Islands, data for which is taken from its Statistics Office website.⁷

We measure GDP using local currency. NPI and NSI data are available from the IMF in current US dollars, which is converted into local currency (for the 9 of the 12 PICs that do not use the US dollar) using the WDI exchange rates⁸. Constant price figures are calculated using a local deflator (with 2015 as the base year) and are then converted to US\$ using the base year's exchange rate.

The starting year of the sample varies by country, depending on the availability and consistency of disaggregated NPI and NSI data. Data is available from the early 1980s for Fiji, Solomon Islands, and Vanuatu and from 2002 onwards for others, with data for all the PICs available from at least 2008, except for the Cook Islands, for which it is available from 2011. NPI and NSI figures are available until 2022 (and for some countries, 2023), except for the Marshall Islands (2021) and Cook Islands

⁶Before renegotiating the Timor-Leste-Australia maritime border, oil production was not included in GDP but instead counted under NPI. This made Timor-Leste's GNDI/GDP ratio the highest in the world: 526 per cent in 2008. In 2021, following the treatment of renegotiation, oil was shifted into GDP.

⁷Available at <https://stats.gov.ck/>.

⁸Note that the Federated States of Micronesia, the Marshall Islands, and Palau, use the US dollar as their local currency; Kiribati, Nauru, and Tuvalu use the Australian dollar (A\$); and the Cook Islands use the New Zealand dollar (NZ\$). The A\$/U\$ exchange rate is applied to Tuvalu.

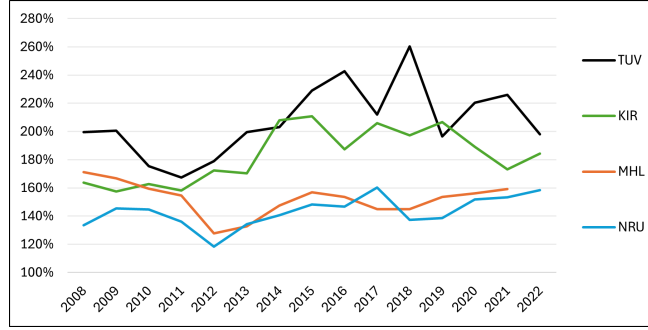
(2019). The GNDI-GDP gap can be decomposed until the most recent year (2022, 2021 or 2019) except for FSM, where data on the components of NPI and NSI are only available from 2009 to 2014.

3 The unique importance of GNDI in the Pacific

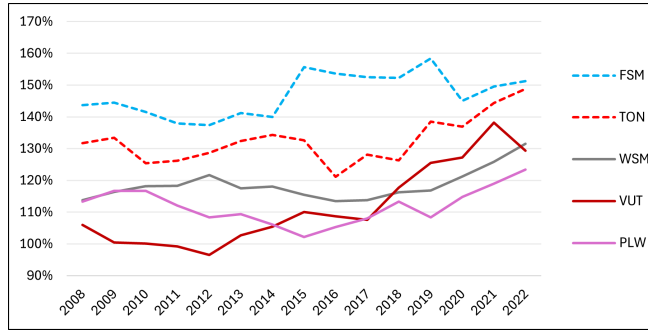
GDP and GNDI are substantially and increasingly different in the Pacific. We can measure the divergence between GNDI and GDP for all PICs from 2008 to 2022, except for the Cook Islands and the Marshall Islands, for which data is available from 2011 to 2019 and 2008 to 2021, respectively.

Figure 2 plots the GNDI/GDP ratio for 12 PICs, dividing them into three groups for legibility. The first group comprises PICs with a very high GNDI/GDP ratio. There are four PICs that, at least by the end of the period, have a GNDI/GDP ratio of around 160 per cent or more: Tuvalu, Kiribati, the Marshall Islands and Nauru. Five PICs have high GNDI/GDP ratios, in the range of around 125-150 per cent by the end of the period: the Federated States of Micronesia, Tonga, Samoa, Vanuatu and Palau. Finally, three PICs have moderate GNDI/GDP ratios of around 105-110 per cent by the end of the period: the Solomon Islands, the Cook Islands and Fiji. Even the last three are above the GNDI/GDP global average. For most of these countries, the trend over the period is upward, though it is never monotonic. The exceptions are the Marshall Islands, the Solomon Islands and Fiji.

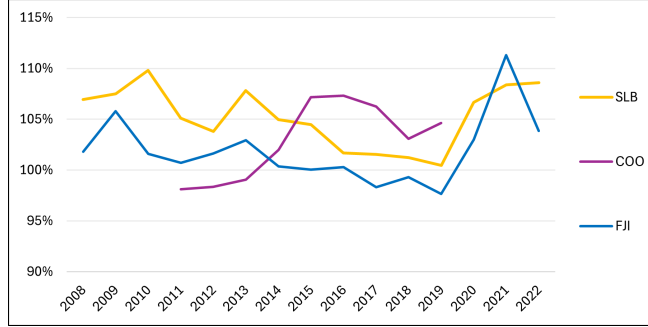
Figure 3 shows the average GNDI/GDP ratio for 12 PICs. In 2008, the ratio was 135 per cent (excluding the Cook Islands, for which data are only available for a shorter period). After falling slightly to just about 125 per cent in 2012, it rose to 140 per cent in 2015 and 145 per cent in 2021.



(a) Very high GNDI/GDP



(b) High GNDI/GDP



(c) Moderate GNDI/GDP

Figure 2: *GNDI-GDP ratio for 12 PICs, 2008-2022.*

Note: Figures for COO and MHL correspond to 2011-2019 and 2008-2021, respectively.

The large and growing gap between GNDI and GDP is not observed in other regions.

Figure 4 plots the GNDI/GDP ratio for all major geographical regions. The annual

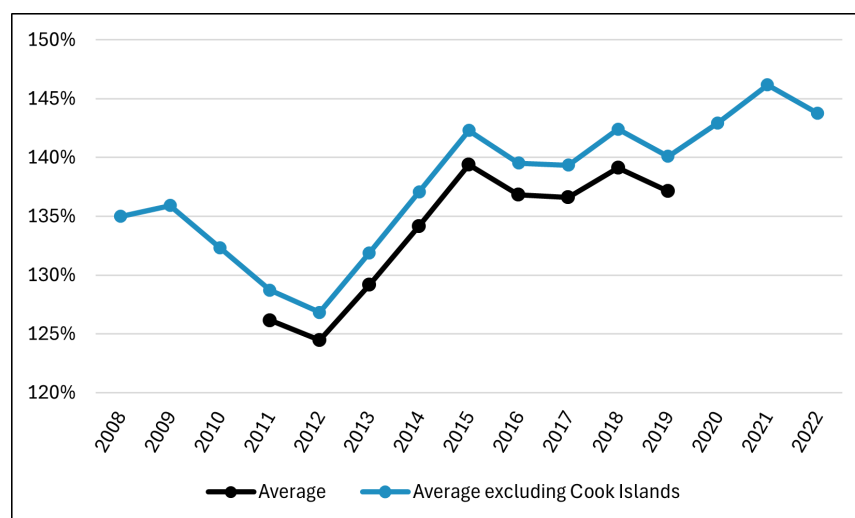


Figure 3: *GNDI-GDP ratio, the average of 12 PICs, 2008-2022.*

Note: The 2022 figure excludes MHL.

average ratio for these regions is narrowly bounded between 95 per cent and 105 per cent. There is no upward trend in any region other than the Pacific.

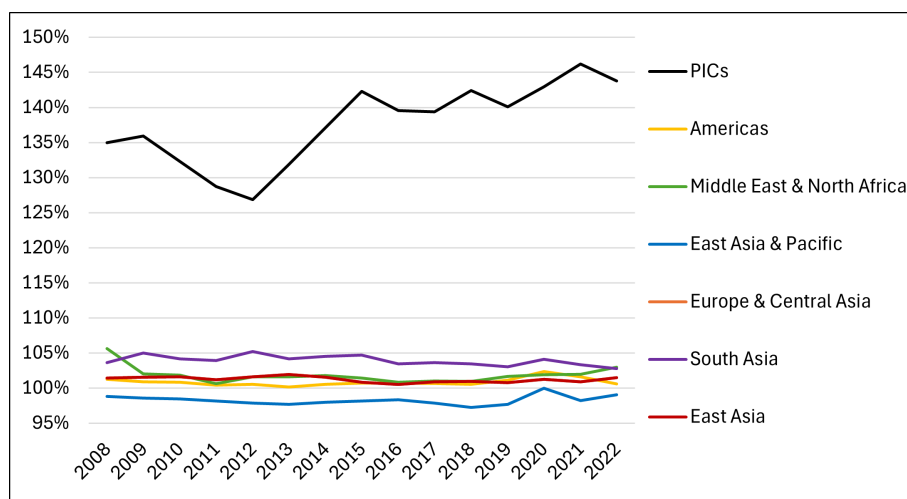


Figure 4: *GNDI-GDP ratio for PICs and other regions, 2008-2022.*

Notes: Timor-Leste is excluded from the East Asia average, and PNG is included. Simple averages are presented. Only countries with available data for every year are included. The Americas comprise countries in North America, Latin America and the Caribbean. For the PICs, the average excluding COO is used.

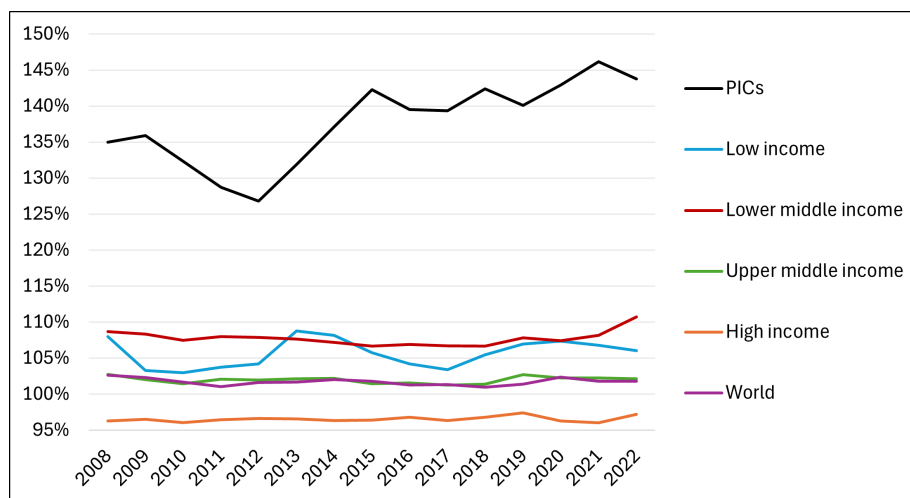


Figure 5: *GNDI-GDP ratio for PICs and other income groupings, 2008-2022.*

Notes: Income groupings and world average exclude PICs. Timor-Leste is also omitted. Only countries with available data for every year are included. For the PICs, the average from Figure 3 (excluding COO) is used.

Three PICs are high-income (the Cook Islands, Nauru and Palau), four are upper-middle-income (Fiji, the Marshall Islands, Tonga and Tuvalu), and five are lower-middle-income (the Federated States of Micronesia, Kiribati, Samoa, the Solomon Islands and Vanuatu). Globally, high-income countries, on average, have a GNDI/GDP ratio of slightly less than one (Figure 5). Middle and low-income countries have a ratio of slightly more than one, as does the world as a whole.

There is no similarity between the PICs and small states in other regions. Figure 6 shows the GNDI/GDP ratio for PICs and small states (population less than 1.5 million) from other regions. The Caribbean is often presented as the region most similar to the Pacific (see, for example, [Armstrong et al., 1998](#)). However, the GNDI/GDP ratio for small states in the Americas (mainly in the Caribbean) is slightly less than one and shows no upward trend. Apart from the PICs, only small states in Africa and the Middle East have a ratio larger than one, and for them, it is just larger than one, and again, there is no upward trend.

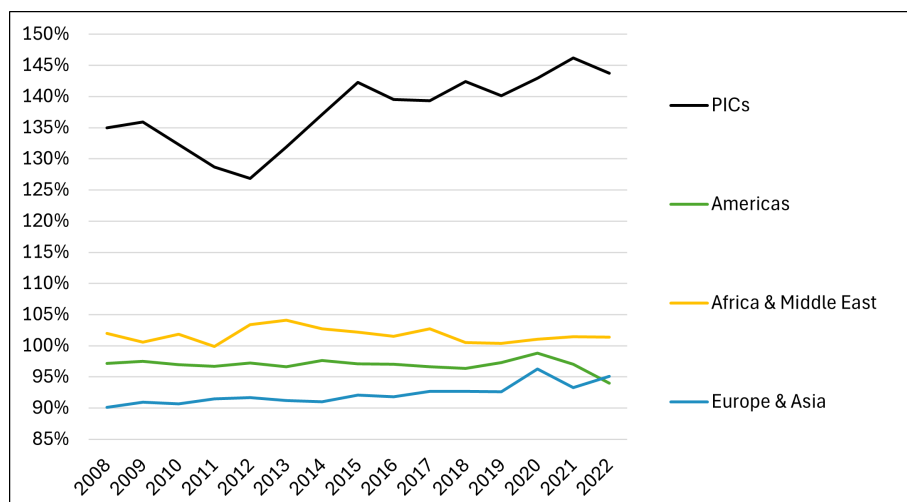


Figure 6: *GNDI-GDP ratio for PICs versus other small state groupings, 2008-2022.*

Notes: Small states are countries with a population of 1.5 million or less. Timor-Leste is omitted. Only countries with available data for every year are included. For the PICs, the average from Figure 3 (excluding COO) is used. There are 11 small states in the Americas: eight from Africa and the Middle East, and nine from Europe and Asia.

While there are individual countries for which the GNDI/GDP ratio is significantly different from one (see Figure 1), the uniqueness of the Pacific island region compared to other groupings, whether of region or income, is evident.

4 Economic performance in the Pacific

Using GNDI or GDP makes an important difference for some PICs when looking at per person income levels (Figure 7). We take 2019 as our comparison year. Using GDP, Tuvalu has an average income for a PIC and is on the threshold of being an upper-middle-income country. Using GNDI, it has an above-average income for a PIC and is halfway between the upper-middle-income and high-income thresholds. Fiji goes in the other direction: from an average to below-average PIC income. Kiribati goes from being the poorest PIC to being significantly better off than the Solomon

Islands and having about the same income per person as Vanuatu. Nauru goes from being middle-income to high-income.

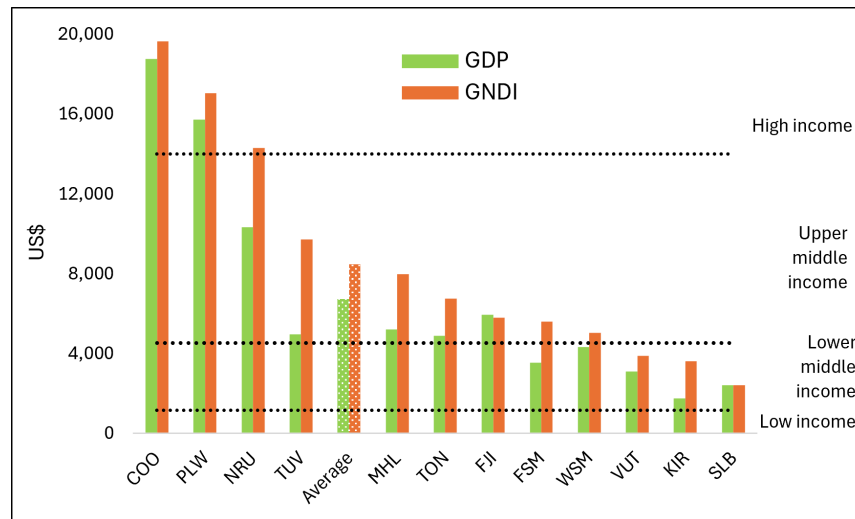


Figure 7: *GNDI and GDP per person for 12 PICs in current US\$, 2019.*

Notes: Countries are ranked based on the size of GNDI. Income thresholds follow the 2025 World Bank [Atlas](#) method.

Using GNDI rather than GDP makes a big difference when it comes to growth rates. This is true regardless of the deflator used.

Figure 8 shows income per person over time using GDP, GNDI and GNDI*, in million US dollars.⁹ There are clear differences in growth rates, especially for Kiribati, Nauru, Tonga, Tuvalu and Vanuatu.

The greater volatility of GNDI is also evident from Figure 8, especially for Kiribati, FSM, Nauru, the Solomon Islands, Tonga and Tuvalu. This is confirmed in Figure 11 in the next section.

Averaging across countries and showing the results in terms of growth rates rather than levels reveals the growth spurt of the 2010s if GNDI is used to measure growth.

⁹Recall from Equations (8) and (9) that GNDI uses CPI as the deflator, while GNDI* uses the GDP deflator.



Figure 8: *Real GDP and GNDI per person, 2008-2022, US\$, 2015 prices.*

Notes: GDP, GNDI and GNDI* are derived from Equations (7), (8) and (9), respectively, and using 2015 exchange rates to convert to US\$. GNDI uses CPI as the deflator, and GNDI* uses the GDP deflator.

This is especially the case if CPI is used as a deflator, but it is also clear, though for a shorter period, if the GDP deflator is used.

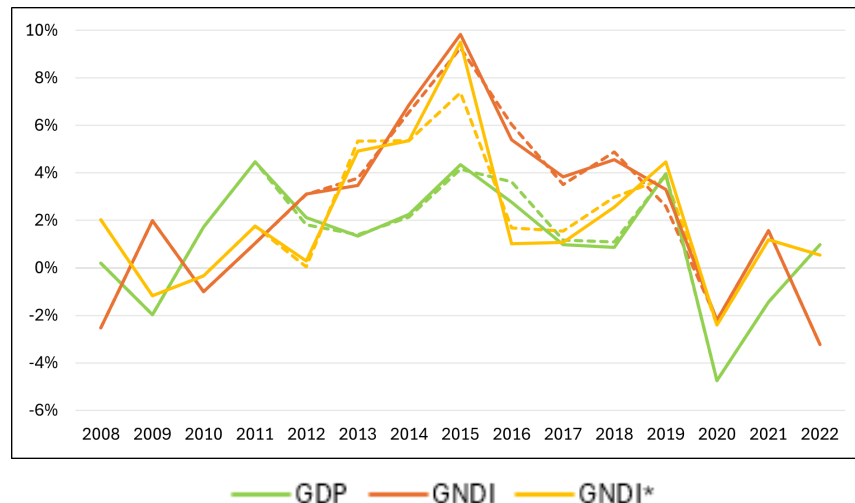


Figure 9: *Average per person growth rates for PICs, 2008-2022.*

Note: Calculated from Figure 8. Data for COO and MHL are only included for 2011-2019 and 2008-2021, respectively. The dashed lines indicate per person growth rates when COO is excluded from the PIC average.

The difference that reliance on GNDI makes to the calculation of growth is even more striking when we look at the Pacific relative to other regions (Table 1). For this comparison, we use average growth from 2011 to 2019¹⁰ (a) to be able to compare a single number across regions and (b) to avoid both the Global Financial Crisis (GFC) and the pandemic.

Using GDP, the Pacific average growth over this period, 2.5 per cent, is average by global standards. It is similar to the average for middle-income countries (2.7%) and above that of high-income countries (1.9%) and low-income countries (0.6%). It is below the average for South Asia (3.8%) and East Asia (3.4%), similar to the average for Europe and Central Asia (2.7%) and above the average for all other regions.

¹⁰That is, the average of 2011 to 2019 growth rates.

Table 1: GDP and GNDI per person growth rates (%) for PICs, by regional and income groupings, average of 2011-2019.

Country/Region/Income group	GDP	GNDI	GNDI*
PICs			
Cooks Islands	1.4	5.9	5.1
Fiji	3.2	3.9	2.7
Federated States of Micronesia	0.2	2.5	1.6
Kiribati	3.6	7.8	6.7
Marshall Islands	4.8	5.1	4.9
Nauru	5.6	10.5	5.1
Palau	2.3	2.1	1.4
Samoa	1.0	0.8	0.9
Solomon Islands	0.9	0.1	0.0
Tonga	2.7	4.3	3.9
Tuvalu	4.3	8.6	6.4
Vanuatu	0.4	4.1	3.0
PICs average	2.5	4.6	3.5
Regional groupings			
World excluding PICs	1.9	2.0	2.0
South Asia	3.7	3.7	3.5
East Asia	3.3	3.1	3.3
Americas	1.5	1.8	1.6
Europe & Central Asia	2.3	2.8	2.3
Middle East & North Africa	1.0	0.6	0.9
Sub-Saharan Africa	1.3	1.1	1.5
Income groupings			
Low income	0.7	-0.4	0.9
Lower-middle income	2.6	3.0	2.5
Upper-middle income	2.4	2.5	2.5
High income	1.5	1.7	1.6
Small states only			
Europe & Asia	1.7	2.2	1.8
Europe & Asia	1.9	2.6	2.2
Americas	0.8	1.4	0.9
Africa & Middle East	3.0	3.1	3.0

Source: [World Bank \(2024\)](#); [IMF \(2024\)](#); authors' calculations.

Note: Growth rate for COO is the average of growth rates from 2011 to 2019 and is included in the Pacific average. GDP, GNDI, and GNDI* growth rates are calculated as per Equations (7), (8) and (9), respectively. See also the notes in Figures 4, 5 and 6.

If we look instead at growth in GNDI, the Pacific 2011-2019 average of 4.6 per cent is well above the average of any other region or income category. The next highest is South Asia (3.7%) and East Asia and the small states of Africa and the Middle

East (both 3.1%).

This Pacific growth spurt is a surprise, especially when set against the Pacific economics literature, which has long portrayed the region as slow-growing. For example, a report from the Asian Development Bank ([ADB, 1999](#)) written at the turn of the century began by noting that the PICs “continue to experience per capita rates of growth significantly below their trading partners and below other small countries elsewhere in the world.” A more recent IMF report talked about the PICs being “stuck in a low-growth path” ([IMF, 2020](#)). As we discuss later, it may be unlikely that the growth experienced by the PICs over the 2010s will be replicated going forward, but at least for that decade, Pacific growth was reasonable, even using GDP and outstanding using GNDI.

The Pacific is the only region where the difference between GNDI and GDP growth over the 2010s is larger than one percentage point in absolute value. In the Pacific, the difference is 2.1 percentage points. The next biggest difference is one-third of this size (0.7 percentage points in the small states of Europe and Asia).

The large difference in GNDI and GDP growth rates in the Pacific is due both to the faster growth in nominal net foreign income than in nominal GDP and to inflation being lower when measured by the CPI than when measured by the GDP deflator. In fact, both are equally influential in the Pacific. About half of the difference in growth rates between GDP and GNDI is due to the faster growth in NPI and NSI, and about half is due to the different deflators. If we deflate GNDI using the GDP deflator (GNDI*), the average Pacific per person growth rate is 3.5 per cent, halfway between the GDP (2.5%) and GNDI (4.6%) per person growth averages.

Some country examples are illuminating. In Vanuatu, looking at GDP per person,

the economy is stagnant, growing at just 0.4 per cent a year on average between 2011 and 2019. But, using GNDI, the economy has been dynamic, growing at an average per person rate of 4.1 per cent. The divergence between GDP and GNDI for the Cook Islands is relatively small. Its GNDI/GDP ratio started off at 97 per cent in 2011 and finished at 105 per cent in 2019 (see Figure 2). Nevertheless, because this ratio, though small in absolute value, is growing, the Cook Islands' GNDI grows much faster than its GDP: 5.9 per cent versus 1.4 per cent on average.

5 Analysis of net foreign income components

This section explains the growing divergence between GDP and GNDI in the Pacific and the recent rapid GNDI growth in the Pacific by examining the components of net foreign income defined in Section 2.2: aid, remittances, investment income, returns on non-financial assets, and the residual.

Figure 10 looks at trends in these five components, showing the average component-to-GDP ratio for all twelve Pacific countries. Up to 2014, the most significant source of difference between GNDI and GDP was foreign aid, which was as high as 18 per cent in 2009. However, aid has fallen relative to GDP and is now below 15 per cent. The return on non-financial assets, principally fishing licenses, has increased from just above 5 per cent until 2010 to as high as 20 per cent in recent years. This, more than anything else, explains the growing divergence between GNDI and GDP in the Pacific. After being flat at around 5 per cent of GDP for much of the period, remittances started to increase in 2016 and reached 10 per cent in 2022, and are now as important as aid. Investment income has fluctuated without a trend, mainly between -1 and 1 per cent, and the residual has remained around 1 per cent.

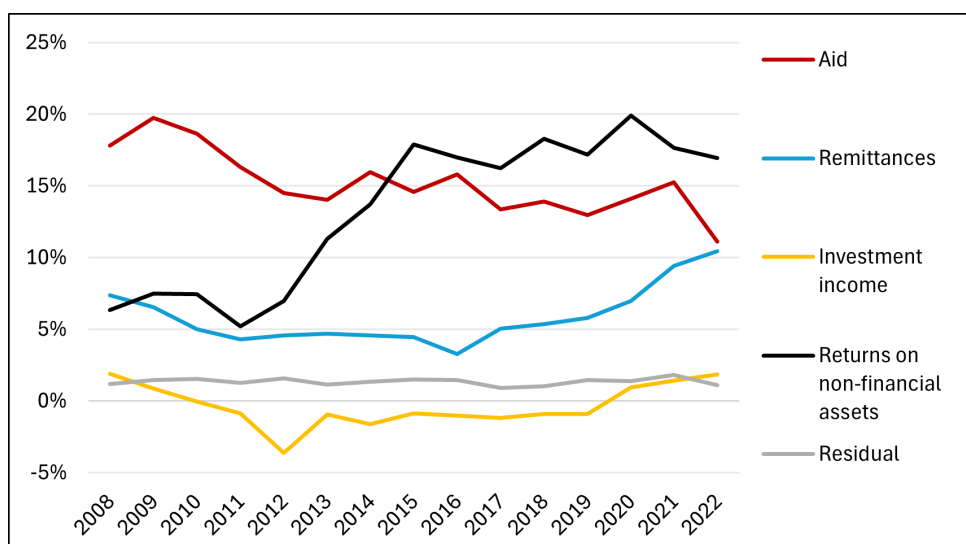


Figure 10: *Trends in the components of net foreign income, average shares of GDP, 2008-2022.*

Note: Data for COO, FSM and MHL are only included for 2011-2019, 2009-2014 and 2008-2021, respectively.

Table 2 shows the components of net foreign income for individual PICs as a share of GDP for the average of the period 2008-2022 (annual data is shown in the appendix, Table A1). Both net foreign income and its components are very unequally distributed across the Pacific. The Gini for average net foreign income as a percentage of GDP is 0.50. It is slightly lower for aid (0.46) but higher for the return on non-financial assets (0.68) and remittances (0.91).¹¹

Given this dispersion, it is difficult to generalise across countries. However, it is notable that the countries with the highest GNDI/GDP ratios tend to have the largest returns on non-financial assets, as well as healthy aid/GDP ratios. The countries with average GNDI/GDP ratios have high levels of remittances.

The rapid growth in per person GNDI in the Pacific over the 2010s can be explained

¹¹All calculations are based on percentages of GDP. Since the average investment income is negative, its Gini is negative, making its comparison with the other components impossible.

Table 2: Components of net foreign income, per cent of GDP, average of 2008-2022.

Country	Aid	Remittances	Investment income	Returns on non-financial assets	Residual	Net foreign income
TUV	43.3	4.8	11.0	47.9	0.2	107.3
KIR	19.7	7.9	12.9	43.0	-0.3	83.2
MHL	31.0	5.9	-6.8	20.0	2.1	52.1
NRU	22.0	-6.7	1.5	26.9	-0.6	43.2
FSM	28.6	1.1	-2.4	8.7	4.4	40.4
TON	3.9	26.4	-0.9	0.0	3.2	32.6
WSM	0.3	18.4	-3.5	0.0	3.3	18.5
PLW	17.1	-4.7	-3.1	2.1	0.4	11.8
VUT	5.9	7.5	-3.4	0.3	1.3	11.7
SLB	8.3	-2.4	-4.0	1.5	1.9	5.3
COO	7.8	0.1	-3.7	-0.5	-0.9	2.9
FJI	1.5	5.5	-5.6	0.0	0.6	1.9

Source: IMF (2024); World Bank (2024).

Notes: Countries are ordered based on the size of their net foreign income (= NPI + NSI) relative to GDP. Figures for COO, FSM and MHL correspond to 2011-2019, 2009-2014 and 2008-2021, respectively.

by a growth decomposition analysis (Table 3), which weights individual component growth rates by their share in GNDI (with all components deflated by the CPI). Almost two-thirds (63 per cent) of the 2011-2019 growth in GNDI per capita is due to GDP growth.¹² More than one-quarter (28 per cent) is due to growth in the return on non-financial assets. Aid and remittances contribute only 4 per cent each. Remittances are growing faster than aid, but aid is more important at the start of the period.¹³

Calculations based on Equation (10) confirm that GNDI is more volatile than GDP, regardless of whether it is deflated by the CPI or the GDP deflator (see Panel (a) of Figure 11). Analysis of the components of net foreign income, estimated from

¹²The average GDP contribution in this table and the average GDP growth rates shown in Figure 9 cannot be directly compared due to the different deflators used: CPI in the former, and GDP in the latter.

¹³Although aid/GDP declined during 2008-2022, it slightly increased between 2011 and 2019.

Table 3: Real GNDI per person growth and the contributions of GDP and net foreign income components, 2008-2022 and 2011-2019 averages.

	GNDI	GDP	NFI	Aid	Remit.	Invest. income	Returns on non-financial assets	Residual
2008	-2.5	-3.1	0.5	0.0	-0.2	0.3	0.6	-0.1
2009	2.0	0.6	1.4	1.1	-0.4	-0.3	1.2	-0.1
2010	-1.0	0.9	-1.9	-0.4	-0.9	-0.6	0.0	0.1
2011	1.0	2.7	-1.7	-0.1	-0.3	-0.5	-0.7	-0.1
2012	3.1	3.7	-0.6	-1.1	0.4	-1.7	1.5	0.3
2013	3.5	-0.1	3.6	-0.7	0.0	2.0	2.6	-0.3
2014	6.9	3.1	3.8	2.2	0.0	-0.6	2.0	0.1
2015	9.8	5.0	4.8	0.5	-0.1	0.1	3.3	0.4
2016	5.4	5.0	0.4	0.8	-0.8	-0.2	0.7	0.0
2017	3.8	2.6	1.2	0.1	1.5	-0.2	0.1	-0.4
2018	4.6	2.0	2.5	0.5	0.3	0.2	1.5	0.1
2019	3.3	2.1	1.2	-0.8	0.7	0.2	0.5	0.4
2020	-2.2	-4.1	1.9	0.6	0.4	1.4	0.8	-0.1
2021	1.6	-1.7	3.3	1.2	1.3	0.6	-0.2	0.3
2022	-3.2	-1.5	-1.7	-1.1	0.4	-0.2	-0.8	-0.5
Average	4.6	2.9	1.7	0.2	0.2	-0.1	1.3	0.1
(% of GNDI growth)	100	63	37	4	4	-2	28	1
Average growth*	4.6	2.5	3.8	0.6	5.9	-0.4	16.2	-0.6

Notes: Net Foreign Income ([NFI](#)); Remit. = Remittances. The table shows the average (of 12 PICs) real GNDI per person growth and the contributions of GDP and NFI components from 2008 to 2022 alongside their average contribution over 2011-2019 (the period analysed in [Table 1](#)). Data for COO and MHL in this calculation are only available for 2011-2019 and 2008-2021, respectively. Since component data for FSM is only available for 2009-2014, the contributions of individual NFI components do not add up to NFI growth in 2008-2009 and 2015 onwards. However, the deviations are very minor and do not change the overall conclusions. The last row of the table shows the average growth rate of GNDI and GDP per person over 2011-2019, calculated as the average of country-specific growth over the same period. For NFI and most of its components, average growth rates are not meaningful due to the presence of extremely high and low growth rates in some years. Therefore, for NFI and its components, the average growth rate is the average of country-specific median annual growth rates. The average medians for GNDI and GDP growth rates are 3.4 and 1.8 per cent, respectively.

the same equation, reveals that investment income is the most unstable component, followed by aid and remittances (see Panel (b) of [Figure 11](#)). Although returns on non-financial assets grew faster than those of any other counterparts (see [Figure](#)

10), they were the least volatile income component over the period 2008-2022. Note that the volatility in the components of net foreign income is much larger than the volatility in their sum (by an order of magnitude).

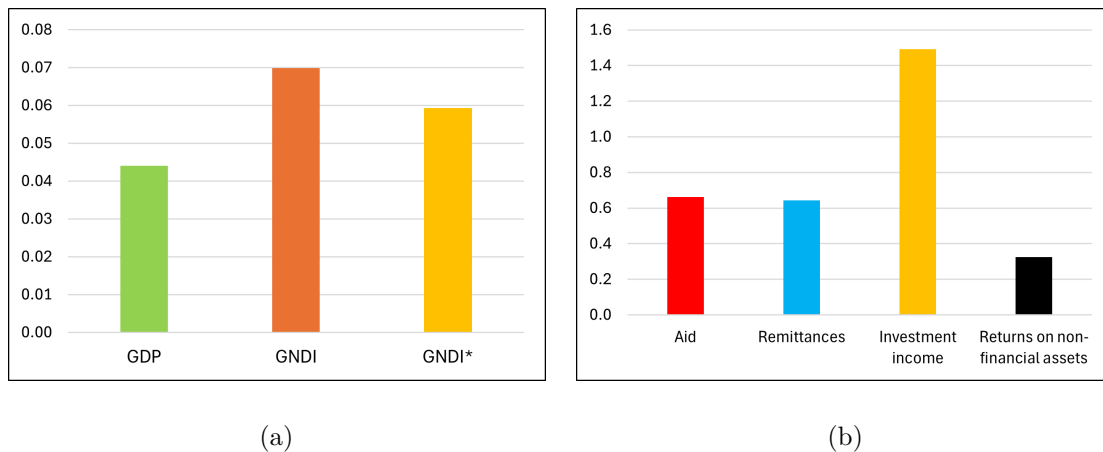


Figure 11: *Volatility in real aggregate national income and the components of net foreign income, 2008-2022.*

Notes: Volatility is estimated from Equation (10), as the average of the standard deviations in the cycle of country-specific (excluding COO and FSM) real income and its components from 2008 to 2022. GDP, GNDI and GNDI* are all deflated as per Equations (7), (8) and (9), respectively.

The following sub-sections explore the components of net foreign income in more detail.

5.1 Aid

Table 4 compares aid as measured by recipients through the balance of payments and aid as measured by donors through data reported to the OECD in 2021. On average, the former is half of the latter. A large difference is expected; as noted earlier, not all aid makes it to the country to which it is allocated, and the aid variable in net foreign income (government transfers) excludes capital transfers (which are recorded in the capital account). However, a few countries have measurement issues.

In Samoa, a negative entry is shown for government transfers, perhaps reflecting interest payments on a concessional loan. In Tonga, only US\$5 million of aid is shown in the balance of payments against recorded aid (from the OECD) of US\$113 million. As noted in the conclusion, more attention to aid reporting in the balance of payments will be crucial if GNDI is to be relied on more in the future.

Table 4: Differences in aid reported by recipients and donors, million US\$, 2021.

Country	BOP	OECD
	government transfers	Official Development Assistance
WSM	-0.4	84.4
TON	5.1	113.1
FJI	218.9	609.4
SLB	98.8	264.7
KIR	44.2	72.9
NRU	21.3	33.8
MHL	73.4	107.1
FSM	81.7	117.0
TUV	29.7	36.0
PLW	44.7	49.4
VUT	154.9	166.9

Source: [World Bank \(2024\)](#) and [IMF \(2024\)](#).

Notes: COO is excluded due to the OECD data unavailability. Countries are ordered according to the difference in aid reported by two sources. Figures for the FSM correspond to 2014.

Overall, aid per person was stagnant during the Pacific growth spurt (see Table 3). The five countries with the highest aid/GDP ratios are Tuvalu, the Marshall Islands, FSM, Kiribati and Palau (see Table 2). Three of these have compacts with the United States, which provide significant budget support. The other two, Tuvalu and Kiribati, receive large amounts of aid from Australia for their size. As Figure 12 shows, aid is volatile, especially for the two countries without compacts. Palau is the only one of the top five to show an upward trend in aid/GDP. This reflects recent

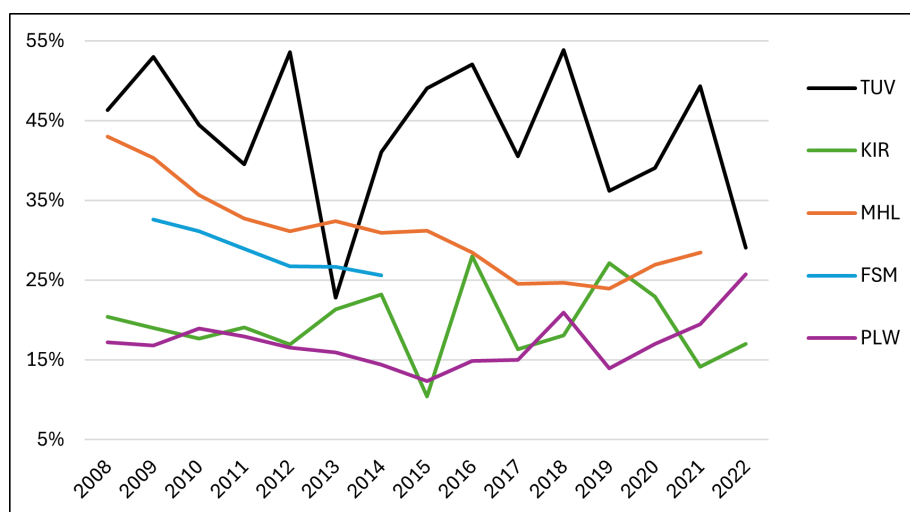


Figure 12: *Trends in aid for the top five PICs, per cent of GDP, 2008-2022.*

Notes: Figures for the FSM and MHL are for the periods 2009-2014 and 2008-2021, respectively.

negative growth in the country's GDP.

5.2 Remittances

As noted in Section 2.2, remittances, as defined by the World Bank, include both compensation of employees (residents in the country but working overseas), which is included in net primary income, and personal transfers, which are included in net secondary income. Most remittances fall in the latter category: for the world, personal transfer receipts are about 70 per cent of remittances received, and for the Pacific, it is 53 per cent (for 2019, excluding FSM).

The inclusion of both types of remittances in GNDI, as opposed to GNI, which only considers the smaller category (compensation of employees), supports focusing on GNDI instead of GNI. The more fundamental argument, however, for preferring GNDI and ignoring GNI is that the division of remittances into these two components

appears to be essentially arbitrary. In general, remittances received exceed remittances sent. However, the sum of the absolute value of the differences (between received and sent) for the two components substantially exceeds the difference between received and sent total remittances, indicating that what one country records as outward compensation of employees is often recorded by another as an inward personal transfer ([Reinke et al., 2007](#)). In 2005, personal transfer credits at US\$151.4 billion were US\$53.9 billion more than personal transfer debits, while employee compensation credits at US\$71.8 billion were US\$10.4 billion less than employee compensation debits. Thus, the sum of the absolute value of the two differences is US\$64.4 billion compared to the actual difference of US\$43.5 billion.

If the global data shows that the distinction between the two types of remittances is unreliable at the global level, an examination of national data shows that it is completely unreliable in the Pacific. According to the IMF guidelines, the income of seasonal workers is to be included as compensation of employees, and their consumption while overseas should be treated as imports. However, there is no obvious way to distinguish remittances sent by seasonal workers from those sent by more permanent migrants.

Empirically, it is evident that countries of the Pacific apply very different methods, some tending to categorise the remittances of seasonal workers as compensation of employees and some as personal transfers ([PACER Plus, 2023](#)). Tonga, Samoa and Vanuatu are all major suppliers of seasonal labour to Australia and New Zealand. However, Samoa classifies *all* remittances as personal transfers; Tonga classifies *nearly all* remittances as personal transfers; and Vanuatu classifies *nearly all* remittances as compensation of employees (Figure 13). The figure also compares FSM and the Marshall Islands, both of which have the same migration rights to

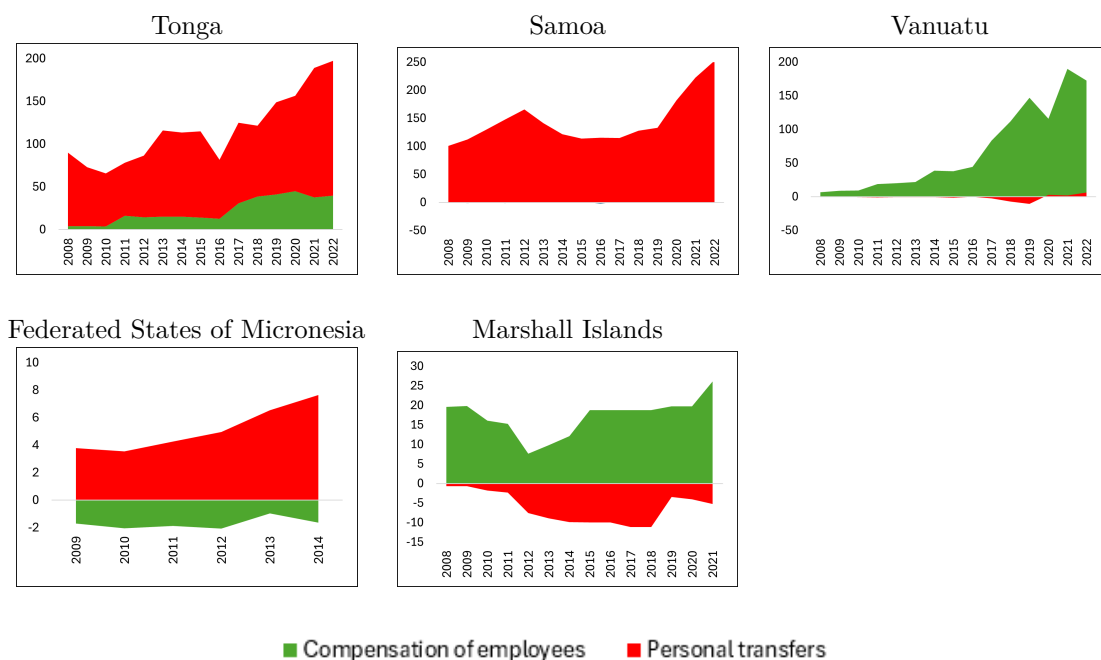


Figure 13: *Composition of net remittances in several Pacific countries, current million US\$, 2008-2022.*

Source: IMF (2024). Note: Data for the FSM and MHL are shown for the years for which they are available, 2009-2014 and 2008-2021, respectively.

the US. The FSM classifies most of its remittances as personal transfers and has a negative compensation of employees. The Marshall Islands does the opposite: it classifies most of its remittances as compensation of employees and has negative personal transfers.

These comparisons reveal that the distinction between the two components of remittances — compensation of employees and personal transfers — is not meaningful, at least in the Pacific. This implies that GNI cannot be relied on since it requires precisely this capacity to distinguish between these two components. This is why, as noted in the introduction, we do not consider GNI as a measure of economic activity or welfare in this paper.

The reliance on remittances among the PICs is highly divergent (see Table 2). For Tonga and Samoa, remittances are as important as aid is to the Pacific’s most aid-dependent countries, constituting 30 per cent or more of GDP. However, at the other end of the scale, remittances are negative for those PICs heavily reliant on expatriate labour: this is the case for Palau and Nauru.

As already noted, remittances have become much more important over the last decade, more or less doubling as a share of GDP (see Figure 10) and becoming as important as aid. However, remittances contributed little to the Pacific growth spurt (only 4 per cent - see Table 3). This is partly because remittances were relatively small at the start of the period and partly because the rapid growth came after 2019 (see Figure 10).

The increased reliance on remittances is by no means uniform. Five PICs have experienced a rise in remittances relative to GDP over the last decade (Figure 14), namely Samoa, Tonga, Vanuatu, Fiji and the Solomon Islands.

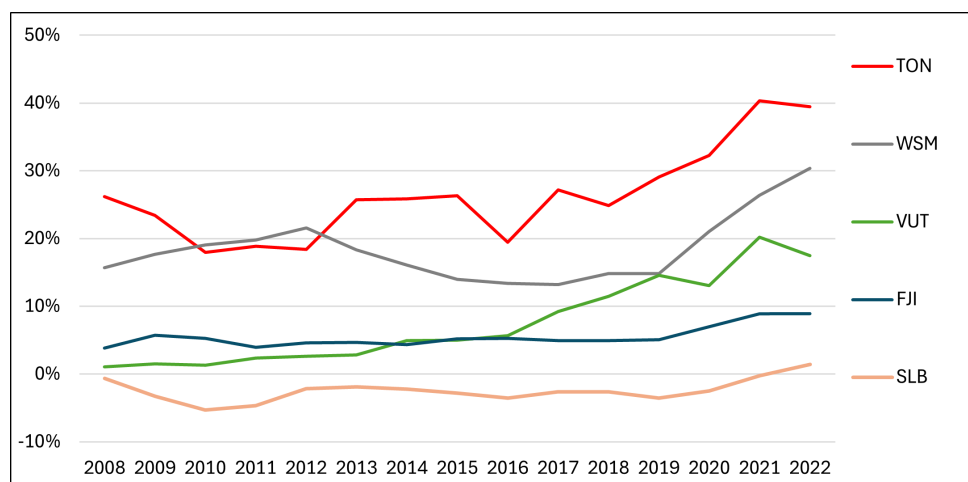


Figure 14: *Positive PIC remittance trends, per cent of GDP, 2008-2022.*

On the other hand, Tuvalu, Palau and Nauru all show a decline in remittances; the

Cook Islands and Kiribati show no or little change, while in the Marshall Islands, remittances have fallen and then increased again (Figure 15).

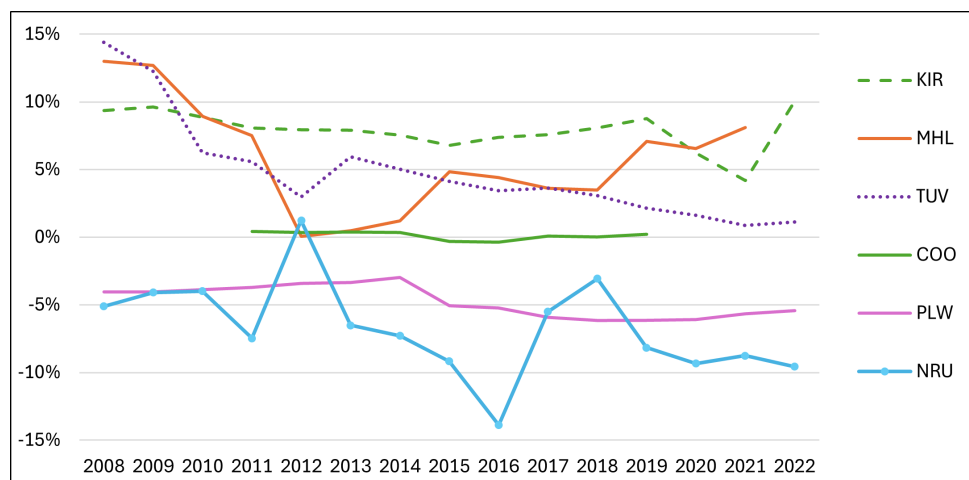


Figure 15: *Negative and neutral PIC remittance trends, per cent of GDP, 2008-2022.*

5.3 Investment income

Investment income is, on average, negative for the Pacific for most of the last decade, and it does not help explain the Pacific growth spurt (see Figure 10 and Table 3).

Trends and levels at the country level are disparate. The most noticeable feature is the difference in levels between Kiribati and Tuvalu on the one hand and the other countries on the other. For 2008-2022, investment income as a percentage of GDP averages 13 per cent for Kiribati and 11 per cent for Tuvalu. Nauru is the only other PIC with a positive average investment income for this period (2% of GDP). All other countries have a small negative average (Fiji's is the largest in absolute value at -6% of GDP).

Investment income is a composite of returns from various asset classes, namely

Foreign Direct Investment (FDI), portfolio investment, central bank reserves and others. Returns on FDI are almost uniformly negative, reflecting the Pacific’s status as a destination country for foreign investment. The only exception is Nauru, which accumulated significant foreign assets in its days of phosphate-extraction-derived wealth. From 2008 to 2010, Nauru’s return on its foreign wealth was 10 per cent of its GDP. However, as Nauru’s economic situation deteriorated, these foreign assets were sold. Its return on FDI went to zero in 2018 and has been negative thereafter. Returns on two other components of investment income, portfolio investments and reserves, are positive for those PICs with a Sovereign Wealth Fund (SWF): Kiribati, Tuvalu, FSM, the Marshall Islands, Palau and Nauru. In Kiribati, the returns on the SWF seem to be shown as a return on reserves, whereas in other countries, they are shown as a return on portfolio investment.

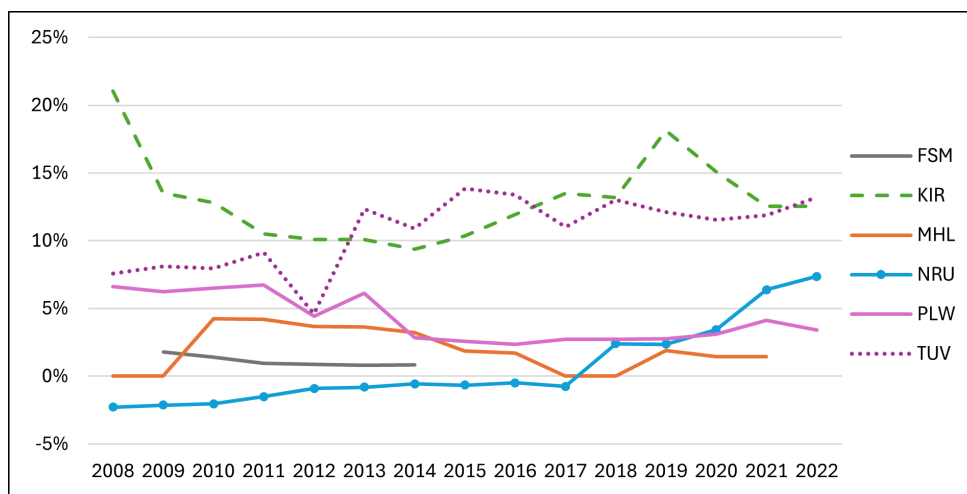


Figure 16: *Returns on portfolio investments and reserves for PICs with a Sovereign Wealth Fund, per cent of GDP, 2008-2022.*

Note: The numerator is the sum of net portfolio investment income and income on reserve assets.

Figure 16 shows the return on portfolio investment and reserves for PICs with a SWF. Kiribati and Tuvalu stand out for the large returns from their SWFs. Tuvalu

and Nauru stand out for growth in their returns. Nauru's SWF is new (established in 2015), and Tuvalu's SWF has grown rapidly.

The return from the SWFs of the three compact states is smaller and typically below 5 per cent. Why this is so requires more research. In 2022, Kiribati's SWF was 306 per cent of its GDP, FSM's 288 per cent, Tuvalu's 225 per cent, the Marshall Islands' 217 per cent, Nauru's 112 per cent, and Palau's 100 per cent.¹⁴ One possibility is a different accounting treatment. If returns from SWFs are reinvested, it is possible that, though they should be, these returns are not credited to net primary income.

5.4 Return on non-financial assets

The large increase in the return on non-financial assets from 5 per cent in 2011 to 18 per cent in 2015 (see Figure 10) is the most important reason for the growing divergence between GNDI and GDP and a key reason for the Pacific growth spurt, explaining 28 per cent of it (see Table 3). This mainly reflects the growth in fishing license revenue due to the Nauru Agreement, which made the PICs akin to a tuna fishing rights cartel (Yeeting et al., 2018). This agreement increased the rents that the Pacific was able to extract from foreign fishing fleets and benefited some countries enormously. The Vessel Day Scheme (VDS) was launched in 2010 and generated US\$64 million in that year. The revenue increased rapidly to reach US\$470 million in 2016 (Aqorau, 2020). A significant portion of this revenue went to PNG, and some of this increase would have been a substitute for previously earned domestic fishing license revenue. Netting these out, the VDS explains the increase in the return on non-financial assets over the same period, which is about US\$200 million.

¹⁴Data from various country-specific Article IV Consultation reports of the IMF.

Income from the VDS has stabilised, which explains why most countries' return on non-financial assets is no longer growing. Nauru is an exception; its return on sovereign assets has continued to increase over the last decade because it derives a return not only from fishing licenses but also from its regional processing centre, which Australia pays for to house (and to potentially house) asylum seekers arriving by boat and other unwanted migrants. Nauru now has the largest return on non-financial assets in the Pacific relative to GDP, 62 per cent in 2022.

We expected that Vanuatu would have shown greater growth in its return on non-financial assets since its citizenship sale program reached over 10 per cent of GDP in 2018 and 2019 ([Surandiran and Howes, 2020](#)). It is unclear where these funds appear in Vanuatu's balance of payments and, indeed, whether they do. It is possible that, since the funds raised from the sale of citizenship are held offshore, they are not included in the BOP until brought onshore. However, since they raise income, these funds should be included in the current account, presumably as a primary income.

Finally, while we analyse the various components separately, there is a link between the growth in this component (return on non-financial assets) and the growth in returns from portfolio investments and reserves discussed in the previous section. Much of the fishing license revenue has been saved into SWFs. Thus expanded, these funds have, in subsequent years, generated greater returns.

5.5 Residual

This component (other net secondary income) is small on average and without trend. One element of it that we can identify is remittances to entities other than households. This is relatively large in countries with large remittances (e.g., Tonga and Samoa)

and would reflect migrant payments to churches.

6 Discussion and conclusion

From a theoretical perspective, GNDI is a better measure of average living standards than GDP. Yet, it is rarely used. This distinction is unimportant to other regions but is very important to the Pacific.

Indeed, the distinction has become more important over time because of the increasing divergence between GNDI and GDP in the Pacific. In 2022, GNDI was, on average, 45 per cent bigger than GDP for the PICs, up from 35 per cent a decade ago. For no income grouping or other region, even a region of small states, the divergence is more than 10 per cent, and in no other region or grouping does this divergence show rapid growth.

Using GDNI when measuring economic growth in the Pacific has several profound implications for our analysis of this region.

First, it challenges the perception of the Pacific as a perpetually slow-growing region. Using GDP for the period between the GFC and the COVID-19 pandemic, the Pacific experienced average growth by global standards. Using GNDI, the Pacific was the fastest-growing region in the world, with an average annual per person growth of 4.6 per cent. This is partly due to the switch in deflators with CPI, which is growing more slowly than the GDP deflator across the Pacific. But even if we remove this effect, by using GNDI with the GDP deflator, the PICs emerge as the equal fastest-growing region in the world over the 2010s.

This remarkable result is unlikely to be repeated in the future. The last decade saw a

massive increase in fishing license revenue flowing to a number of PICs. One-quarter of the average growth in GNDI between the GFC and the pandemic is due to growth in non-financial assets, of which fishing license revenue is the major contributor. This boosted GNDI directly and likely had an indirect, stimulatory effect on GDP. Between 2012 and 2015, the return on non-financial assets increased from 7 per cent of GDP to 17.9 per cent of GDP. It is surely not a coincidence that GDP per person growth peaked in 2015 at above 4.0 per cent (see Figure 9). Fishing license revenue cannot continue to grow indefinitely and, in fact, has already plateaued.

It is, of course, true that the Pacific faces various growth challenges, but in a region that is so uniquely reliant on foreign income flows, growth spurts and dips may well be missed unless the right variable is chosen to measure economic growth. Indeed, the failure of GDP to pick up this growth spurt is reason enough in and of itself to measure economic performance in the Pacific using not GDP but GNDI.

Second, using GNDI changes our perception of individual PICs. Nauru, Tuvalu and Kiribati emerge as economic superstars over the 2010s, with the 2011-2019 average per person growth of 10.5, 8.6 and 7.8 per cent, respectively. Vanuatu appears as a stagnant economy measured using GDP but as a dynamic one measured with GNDI, with an average GNDI per person growth of 4.1 per cent compared to GDP growth of only 0.4 per cent.

Third, using GNDI resolves the long-running debate about the applicability of the MIRAB model to the Pacific. The model correctly suggests that the Pacific is unique in its economic structure and that GDP is a misleading indicator for this region. However, focusing only on aid and remittances would be misleading. As important as these are, the return from non-financial assets is, on average, higher than either,

and in recent years, has grown much faster than both of them.¹⁵ More broadly, using GNDI also provides a framework within which the various contributions to income, domestic and foreign, can be integrated and compared, and debates about the importance of GDP, aid, remittances and other sources of foreign income clarified, if not resolved.

In addition to arguing that GNDI should be, we have also argued that GNI should not be used as a measure of economic activity. GNI is limited in its value because it excludes most remittances and foreign aid. However, it is also conceptually flawed because the distinction between employment compensation (in GNI) and migrant transfers (not in GNI) is unreliable, especially in the Pacific.

In summary, the case for focusing on GNDI when analysing economic performance in the Pacific is overwhelming. Several PICs present GNDI annually, but neither they nor any of the external analysts of the region, such as the multilateral banks, analyse GNDI or look at real GNDI growth rates.¹⁶ They should. No doubt, the estimation of GNDI includes measurement error. However, so does the measurement of GDP, and when foreign income is as important as it is in the Pacific, it is much better to include an imperfect measure of it than to exclude it altogether.

In addition to this clear conclusion, this study also throws up both research questions and data needs. One key question is why the Pacific is so different from other regions in terms of the magnitude of the divergence between GNDI and GDP. The

¹⁵To be fair to the MIRAB theorists, they themselves recognise the importance of non-aid, non-remittance sources of foreign income. [Bertram \(1986\)](#) wrote: “[A]id, philately and migrant remittances are not merely supplements to local incomes, they are the foundations of the modern economy.” This reflected a time in which some Pacific countries sold national postage stamps as a source of income. However, the common understanding of MIRAB, as well as the very term itself, puts the spotlight squarely on aid and remittances to the exclusion of other sources.

¹⁶Five PICs report GNDI: the Federated States of Micronesia, Kiribati, the Marshall Islands, Tonga and Vanuatu. Fiji reports net national disposable income.

explanation, surprisingly, does not relate to the population size (see Figure 6). Part of the answer is obvious. The Pacific is extremely aid-dependent. In 2019, five of the top seven countries in the world in terms of ODA/GDP were all from the Pacific, as were nine of the top 16. The Pacific is also relatively remittance-dependent. In 2019, three (Tonga, Samoa and Vanuatu) of the top 15 countries in the world in terms of net remittances/GDP were from the Pacific.

For a more comprehensive explanation, however, the same sort of analysis we have undertaken for the Pacific in this paper needs to be undertaken for other regions. Our hypothesis is that no other region has the combined characteristics of large aid inflows, remittances, and returns on non-financial assets, as well as relatively low FDI. We venture that it is the combination of these features that makes the Pacific island region unique when it comes to reliance on foreign income.

Data deficiencies need to be overcome in some Pacific island countries for more accurate and timely GNDI reporting. First, for some countries, there is clear evidence of under-reporting of foreign aid in the balance of payments. Second, for Vanuatu, it is unclear where revenue from the sale of citizenship is captured. Third, more recent data needs to be made available by some countries, especially the Cook Islands.

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A Appendix

Table A1: Summary of national income and net foreign income components, average of 2008-2022, million US\$ and per cent of GDP.

Country	GDP	GNDI	NFI	Aid	Remittances	Investment income	Returns on non-financial assets	Residual
COO	320	330	10	25	0	-12	-1	-3
		(103.1%)	(3.1%)	(7.9%)	(0.1%)	(-3.7%)	(-0.3%)	(-0.8%)
FJI	4403	4475	72	66	244	-260	0	22
		(101.6%)	(1.6%)	(1.5%)	(5.5%)	(-5.9%)	(0.0%)	(0.5%)
FSM	340	501	161	88	3	-8	27	14
		(147.4%)	(47.4%)	(25.9%)	(1.0%)	(-2.2%)	(8.0%)	(4.0%)
KIR	207	383	175	41	16	26	92	-1
		(184.4%)	(84.4%)	(19.9%)	(7.8%)	(12.8%)	(44.3%)	(-0.3%)
MHL	195	296	101	59	11	-13	40	4
		(151.7%)	(51.7%)	(30.2%)	(5.7%)	(-6.8%)	(20.5%)	(2.1%)
NRU	99	144	44	19	-7	1	32	-1
		(144.7%)	(44.7%)	(19.1%)	(-7.1%)	(1.5%)	(31.8%)	(-0.7%)
PLW	243	271	28	41	-12	-9	6	1
		(111.4%)	(11.4%)	(17.0%)	(-4.9%)	(-3.6%)	(2.3%)	(0.5%)
SLB	1291	1355	64	95	-30	-45	20	24
		(105.0%)	(5.0%)	(7.4%)	(-2.3%)	(-3.5%)	(1.6%)	(1.9%)
TON	438	583	144	17	117	-4	0	14
		(132.9%)	(32.9%)	(4.0%)	(26.8%)	(-1.0%)	(0.0%)	(3.2%)
TUV	43	90	47	18	2	5	22	0
		(209.0%)	(109.0%)	(42.8%)	(4.2%)	(11.3%)	(50.5%)	(0.2%)
VUT	802	908	106	51	68	-26	3	11
		(113.2%)	(13.2%)	(6.3%)	(8.4%)	(-3.3%)	(0.3%)	(1.4%)
WSM	797	945	148	2	147	-27	0	27
		(118.6%)	(18.6%)	(0.3%)	(18.4%)	(-3.4%)	(0.0%)	(3.3%)

Note: Countries are ordered alphabetically. NFI = Net foreign income. Figures for COO and MHL correspond to 2011-2019 and 2008-2021, respectively. Data for the components of NFI for FSM is available only from 2009 to 2014, and hence, these components do not add up to the NFI. Figures in the parentheses are shown in per cent of GDP.

Table A2: Unit root test results for the cycles of national income and the components of net foreign income, 2008-2022.

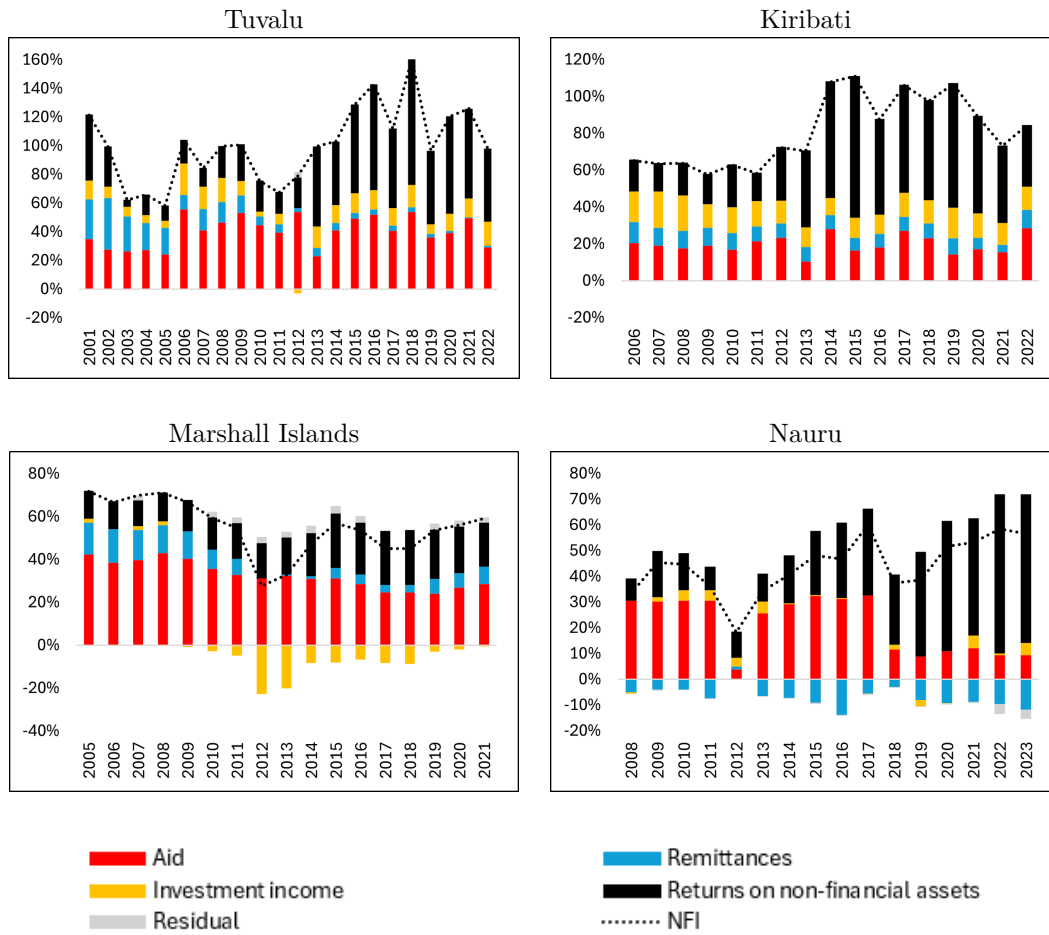
	ADF	PP	KPSS
	p-value		LM-statistic
Fiji			
Aid	0.005	0.005	0.175
Remittances	0.000	0.040	0.299
Investment income	0.006	0.006	0.086
Return on non-financial assets	...	...	...
GDP	0.015	0.141	0.076
GNDI	0.063	0.217	0.088
GNDI*	0.012	0.227	0.074
Kiribati			
Aid	0.003	0.000	0.423
Remittances	0.294	0.013	0.410
Investment income	0.239	0.053	0.098
Return on non-financial assets	0.029	0.050	0.081
GDP	0.139	0.130	0.098
GNDI	0.051	0.049	0.099
GNDI*	0.084	0.082	0.095
Marshall Islands			
Aid	0.219	0.192	0.073
Remittances	0.107	0.110	0.072
Investment income	0.008	0.014	0.103
Return on non-financial assets	0.150	0.402	0.097
GDP	0.108	0.123	0.095
GNDI	0.078	0.219	0.068
GNDI*	0.022	0.082	0.301
Nauru			
Aid	0.036	0.037	0.065
Remittances	0.008	0.006	0.134
Investment income	0.016	0.016	0.096
Return on non-financial assets	0.053	0.025	0.090
GDP	0.157	0.126	0.099
GNDI	0.001	0.000	0.264
GNDI*	0.023	0.249	0.069
Palau			
Aid	0.026	0.013	0.056
Remittances	0.437	0.370	0.090
Investment income	0.045	0.049	0.319
Return on non-financial assets	0.053	0.053	0.103
GDP	0.116	0.097	0.137
GNDI	0.077	0.275	0.092
GNDI*	0.202	0.186	0.103

Continued ...

	ADF	PP	KPSS
	p-value		LM-statistic
Solomon Islands			
Aid	0.016	0.088	0.085
Remittances	0.013	0.630	0.115
Investment income	0.015	0.001	0.433
Return on non-financial assets	0.007	0.001	0.500
GDP	0.007	0.035	0.247
GNDI	0.005	0.121	0.092
GNDI*	0.104	0.109	0.120
Samoa			
Aid	0.081	0.089	0.112
Remittances	0.237	0.205	0.088
Investment income	0.036	0.035	0.084
Return on non-financial assets	0.011	0.000	0.500
GDP	0.037	0.525	0.095
GNDI	0.219	0.382	0.102
GNDI*	0.031	0.262	0.103
Tonga			
Aid	0.031	0.206	0.074
Remittances	0.015	0.022	0.070
Investment income	0.096	0.106	0.085
Return on non-financial assets	0.034	0.035	0.090
GDP	0.151	0.118	0.103
GNDI	0.121	0.121	0.083
GNDI*	0.019	0.019	0.108
Tuvalu			
Aid	0.033	0.033	0.090
Remittances	0.130	0.132	0.098
Investment income	0.010	0.010	0.082
Return on non-financial assets	0.105	0.105	0.115
GDP	0.161	0.148	0.108
GNDI	0.019	0.221	0.096
GNDI*	0.006	0.163	0.107
Vanuatu			
Aid	0.219	0.219	0.099
Remittances	0.014	0.014	0.090
Investment income	0.130	0.130	0.083
Return on non-financial assets	0.008	0.057	0.172
GDP	0.108	0.420	0.088
GNDI	0.345	0.295	0.105
GNDI*	0.311	0.285	0.099

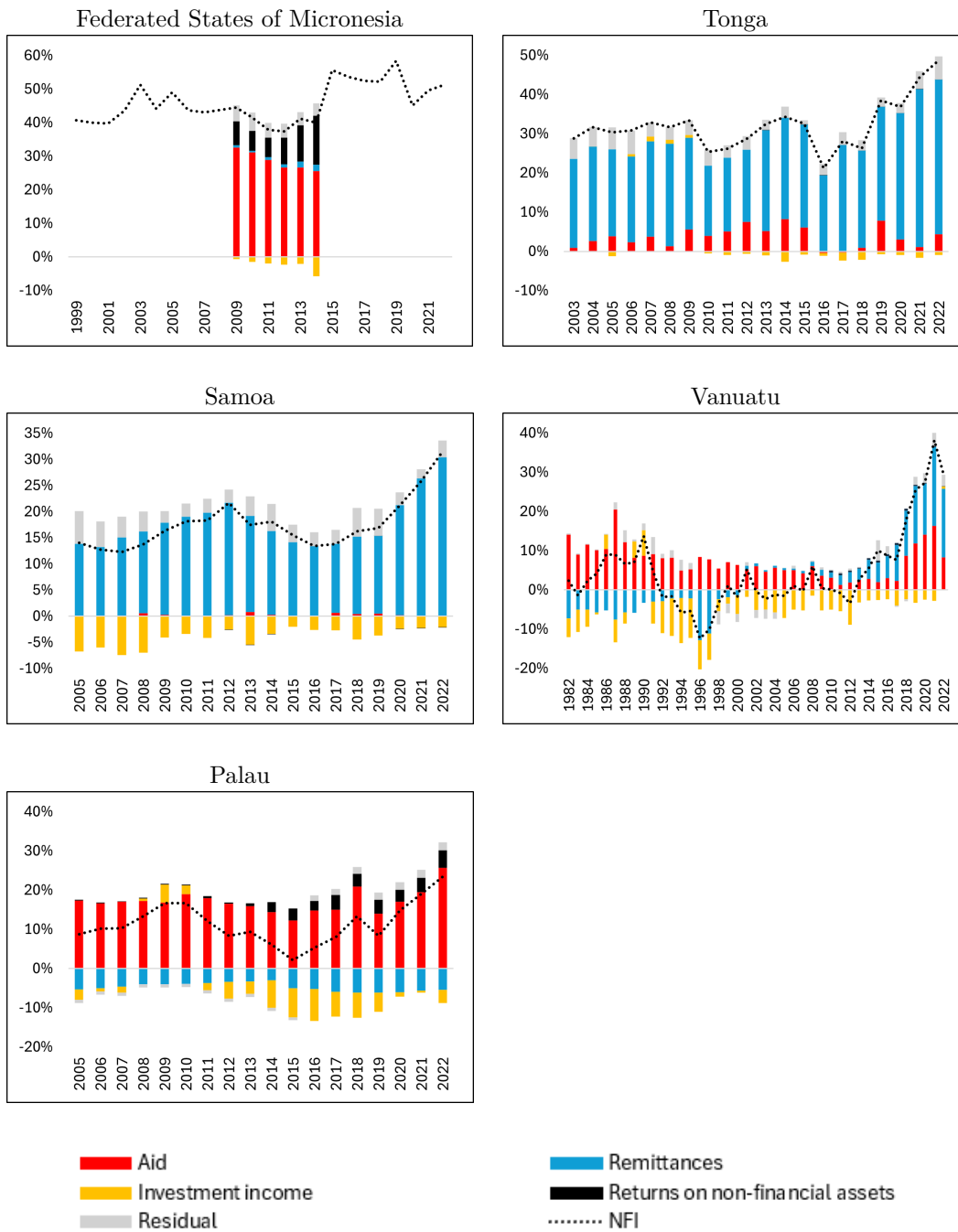
Notes: The null hypothesis (H_0) for the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests is that the variable is non-stationary. A p-value less than 0.05 indicates rejection of H_0 at the 5% significance level. For the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, the asymptotic critical values at the 1%, 5%, and 10% significance levels are 0.739, 0.463, and 0.347, respectively. An LM statistic that exceeds the critical value indicates rejection of the null hypothesis that the underlying series is stationary at the corresponding significance level. Cells labelled ‘...’ indicate that the value of the series is zero for most of the sample period, impacting the estimation of the standard deviation.

(a) Very high NFI/GDP



Continued ...

(b) High NFI/GDP



Continued ...

(c) Moderate NFI/GDP

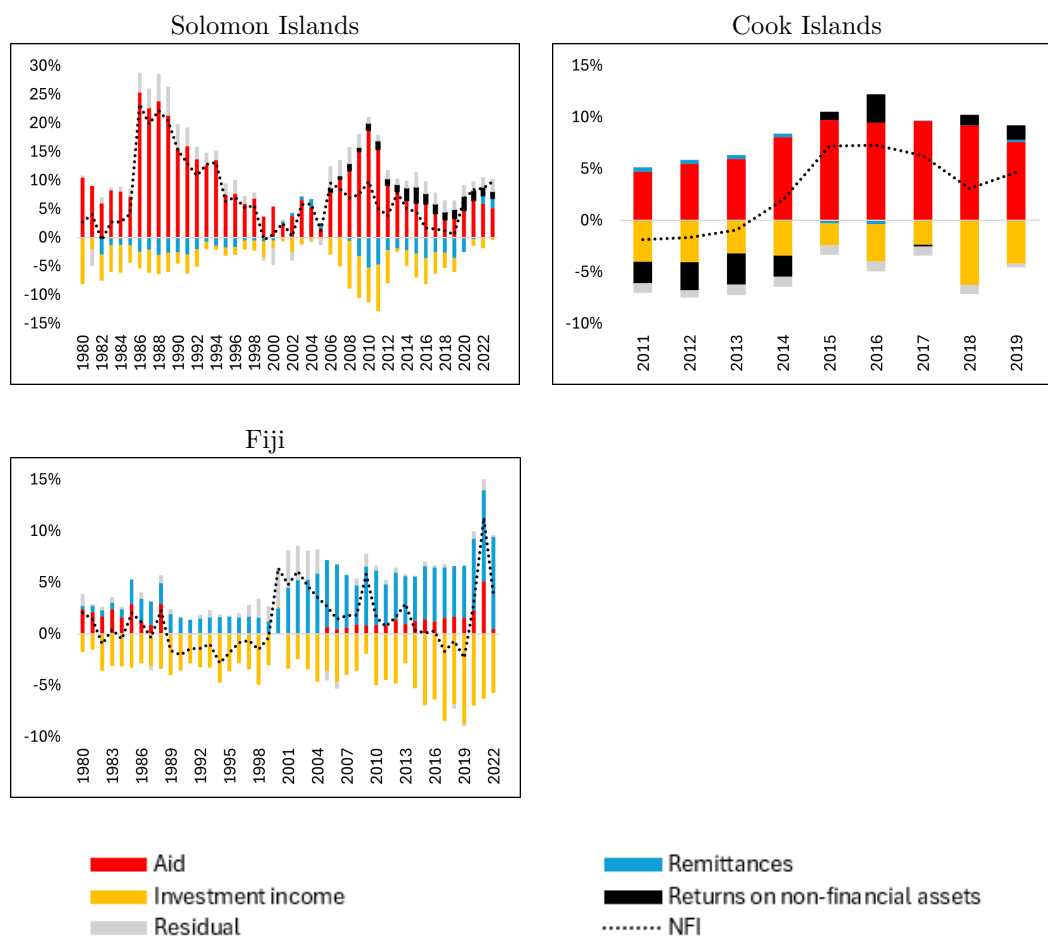


Figure A1: *Net foreign income and its components, per cent of GDP.*

Source: [World Bank \(2024\)](#); [IMF \(2024\)](#); authors' calculation.

Notes: Sample period varies by country depending on data availability. Components of NFI for the FSM are only available from 2009 to 2014.